

Benchmarking Classical Grid-Based Planners in ROS2

Mohamed Eljahmi
RBE550 – October 6 Status

Motivation & Goals

- Path planning = core problem in robotics
- Compare classical planners under identical conditions
- Planned algorithms: BFS, Dijkstra, A*, Theta*, JPS

Framework Overview

- ROS2 + Python implementation
- Unified planner interface (`plan()` $\rightarrow$ `PlanResult`)
- Utilities for maps, neighbors, metrics

Implemented So Far

- BFS: exhaustive, guarantees shortest path
- A*: heuristic-guided, Manhattan & Euclidean
- Run scripts: reproducible experiments

Benchmark Setup

- Random grid: 64×64 , 20% obstacles
- ASCII map: maze_32.txt
- Metrics: path length, runtime, nodes expanded, turns

Preliminary Results

Algorithm	Connectivity	Heuristic	Path Length	Nodes Expanded	Runtime (ms)
BFS	4	-	64	2155	2.39
A*	4	Manhattan	64	910	1.93
A*	8	Euclidean	45	455	1.81

- A* expands fewer nodes
- Heuristics reduce exploration
- 8-connected + Euclidean shortens path

Challenges

- Only single seeded runs so far
- Visualization & plotting pending
- Additional algorithms not benchmarked yet

Next steps

- Benchmark Dijkstra (already coded)
- Implement Theta* and JPS
- Aggregate results, add plots

Packaging & Reproducibility

- Deliver full ROS2 workspace (ros2projrbe550_ws)
- Exclude build/install/log
- Scripts: build.sh, run.sh, run_random64.sh, run_maze32.sh

Conclusion

- Framework operational in ROS2
- BFS vs A* results confirm trade-offs
- Extensible to more planners